Deliverable 2: UML Diagrams Use-Cases, and Scenarios

CCHive: Project Report

CSCI 275

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Table of Contents

1 Introduction 4

2 Current System 4

3 Proposed System 5

3.1 Overview 5

3.2 Functional Requirements 5

3.3 Nonfunctional Requirements 6

4 System Models 7

4.1 Scenarios 7

4.1.1 Scenario vs. Requirements Matrix 9

4.2 Use Case Model 10

4.2.1 Matrix of Requirements vs. Use Cases 14

5 Meeting Details 14

6 Team Contributions 15

1 Introduction

The CCHive project is an initiative to build a student community website that connects learners for the purpose of buying/selling study materials, forming study groups, seeking or offering peer tutoring, and accessing course/instructor reviews. This report outlines the current situation, the problems faced by students, and the proposed solution complete with requirements, constraints, scenarios, and use case models.

2 Current System

2.1 CCHive: The Current Situation

The internet has enabled virtual communities in various forms. However, on many college campuses, there is no unified platform for students to:

- Efficiently buy or sell textbooks/notes that are no longer needed.
- Form study groups and connect with like-minded peers in real time.
- Offer or seek peer tutoring in a convenient manner.
- Rate courses and instructors, helping new students make informed decisions.

Currently, students rely on fragmented methods such as bulletin boards, word-of-mouth, or third-party social media groups. Each approach comes with its own drawbacks:

- **Environmental and economic inefficiency:** Excessive printing of new textbooks when many used copies are available.
- **Limited visibility:** Students interested in voluntary peer tutoring struggle to reach enough potential tutees.
- **No unified review system:** Students have difficulty checking course/instructor feedback before choosing a class.

2.2 The Problems to be solved

1. **Unused Books:** Students completing a course end up with textbooks they no longer need. New students, on the other hand, often must purchase brand-new books at high cost. There is no convenient, centralized platform to facilitate the exchange or sale of used textbooks and materials.

2. **Lack of Structured Study Groups:** Students who want to find peers studying the same course or subject have no single reliable system to connect and form or join study groups.

3. **Peer Tutoring Gap:** Upper-year or more advanced students who are willing to volunteer as tutors cannot easily advertise their help and manage appointments.

4. **Inconsistent Course & Instructor Reviews:** Students cannot effectively check real reviews or feedback about a course or instructor, making it harder to choose classes that fit their learning preferences.

3 Proposed System

3.1 Overview

The proposed CCHive website addresses the gaps identified in the current situation:

1. **Marketplace for Books:** Students will list and discover used books or study materials.

2. **Study Group Platform:** Users can form or join groups, share files, schedule meetups, and receive notifications.

3. **Peer Tutoring Module**: A volunteer tutoring system with scheduling, chat, and feedback loops.

4. **Course & Instructor Review:** Ratings and comments to guide students' course selection.

3.2 Functional Requirements

Each requirement is labeled as **REQ-F#** for reference:

- **REQ-F1: User Registration & Login**

  Students must be able to create an account using their institutional email.

  Authentication via email verification.

- **REQ-F2: List Books**

  Users can create listings with details (title, condition, price, or exchange terms).

- **REQ-F3: Sort and Filter Books**

  Sort listings by price, relevance, or course code.

  Filter by condition, price range, or subject.

- **REQ-F4: Check Prices Without Login**

Visitors can browse and see book prices without creating an account.

- **REQ-F5: Chat with Seller**

  Logged-in users can send direct messages to sellers to negotiate or finalize sale details.

- **REQ-F6: Reviews and Feedback for Sellers**

  Buyers can leave feedback on sellers post-transaction.

- **REQ-F7: Create and Join Study Groups**

  Users can create a new study group or join existing ones based on course or topic.

- **REQ-F8: Share Materials in Groups**

  Group members can share notes, links, or other resources.

- **REQ-F9: Notification of Group Meetings**

  System sends email or app notifications about upcoming group study sessions.

- **REQ-F10: Check Course/Instructor Reviews Without Login**

  Visitors can read ratings and feedback on courses and instructors to make informed decisions.

- **REQ-F11: Peer Tutoring Registration**

  Advanced students can register as tutors, specifying their availability, subjects, and rates (if any).

- **REQ-F12: Book Tutor Appointments**

  Students can select tutors, check their schedules, and request an appointment slot.

3.3 Nonfunctional Requirements

Each requirement is labeled **REQ-NF#** for reference:

1. **REQ-NF1:** The interface should allow a new user to post a book listing within 5 minutes of first interaction.

2. **REQ-NF2:** Provide a responsive design that adapts to mobile, tablet, and desktop.

3. **REQ-NF3:** The system must be available 24/7 with a maximum planned downtime of 1 hour/month.

4. **REQ-NF4:** Automated backups should run daily to prevent data loss in case of system failure.

5. **REQ-NF5:** Page load times for the marketplace must not exceed 2 seconds under normal loads.
6. **REQ-NF6:** Chat messages should appear in under 1 second after sending.
7. **REQ-NF7:** All user data (passwords, personal info) must be encrypted in transit (HTTPS).
8. **REQ-NF8:** Sellers with consistently low ratings may be flagged for review to maintain a safe trading environment.
9. **REQ-NF9:** The system should handle unexpected user inputs gracefully and provide meaningful error messages.
10. **REQ-NF10:** Must support up to 10,000 concurrent users for peak enrollment periods.

3.4 Constraints ("Pseudo Requirements")

- **Language & Framework Constraint:** Must use Ruby on Rails for the backend and React for the frontend.
- **Delivery Deadline:** Must be completed and demonstrated by end of semester.

4 System Models

4.1 Scenarios

Each scenario is labeled **SC-x** for reference.

SC-1: Selling a Book

- Bob, a third-year student, logs into CCHive.
- He navigates to the Marketplace and clicks "Create Listing."
- Bob enters the book title, course relevance, price, and condition.
- The system notifies Bob when another student expresses interest.
- Bob arranges a meetup time with the buyer via the chat system, completes the exchange, and marks the listing "sold."
- The buyer later leaves a review for Bob.

SC-2: Buying a Book

- Alice, a first-year student, visits the CCHive homepage without logging in.
- She filters book listings by course code and sees Bob's listing.
- She logs in to chat with Bob about price negotiation.
- Once they agree, they meet on campus to finalize the transaction.
- After the sale, Alice rates Bob's reliability as a seller.

SC-3: Professor Review

- Charlie needs to choose between two instructors for the same course.
- Charlie logs into CCHive, goes to the "Reviews" section, and searches the course code.
- He reads existing reviews for each instructor, comparing teaching styles and difficulty.
- Based on the feedback, Charlie selects the instructor who best matches his learning preference.

SC-4: In Need of Academic Help

- Dana, who recently missed classes, logs in to request a tutor for Subject X.
- She posts a help request specifying the topic and her availability.
- Eve, a senior tutor specializing in that subject, sees the request and offers help.
- Dana schedules an appointment with Eve through the system.
- After the session, Dana marks the request as "Fulfilled" and leaves feedback for Eve.

SC-5: Creating a Study Group

- Frank is preparing for a major exam and wants to study with peers.
- He creates a "Study Group" under the relevant course category.
- The system broadcasts an invitation to interested students.
- Frank schedules a group meeting time; notifications go out to group members.
- Group members share resources and tips in the integrated chat forum.

SC-6: Leaving a Detailed Course Review

- *Grace*, who just completed a challenging course, wants to provide feedback to help other students.
- She logs in to CCHive and navigates to the "Rate Your Course" section.

- She selects the course she recently completed and the instructor.
- Grace gives a star rating and writes a detailed review about the instructor's teaching style, grading policies, and overall difficulty.
- The system adds her review to the database and confirms the submission.
- Future visitors to the course page can see Grace's review and consider it when deciding their own enrollment.

4.1.1 Scenario vs. Requirements Matrix

Scenario \ Requirement	**REQ-F2(List Books)**	**REQ-F3 (Sort/Filter)**	**REQ-F5 (Chat Seller)**	**REQ-F7 (Create/Join Group)**	**REQ-F11 (Tutor Reg.)**	**REQ-F12 (Book Appt.)**	**REQ-F9 (Notify User)**
SC-1 (Selling a Book)	✓	-	✓	-	-	-	✓
SC-2 (Buying a Book)	-	✓	✓	-	-	-	✓
SC-3 (Professor Rev.)	-	-	-	-	-	-	-
SC-4 (Academic Help)	-	-	-	-	✓	✓	-
SC-5 (Study Group)	-	-	-	✓	-	-	✓
SC-6 (Course Review)	-	-	-	-	-	-	✓

4.2 Use Case Model

Each use case references the participating actors, flow of events, entry/exit conditions, and special requirements.

Use Case #1: Sell or Exchange a Book

Use Case Name: Sell or Exchange a Book

Participating Actors:

- Seller (Student who wants to post a listing)
- System (CCHive website)

Entry Condition:

- The Seller is logged into CCHive.

Flow of Events:

- The Seller navigates to the “Marketplace” section and selects “Create Listing.”
- The System presents a form for book details (title, course code, price/exchange terms, etc.).
- The Seller submits the form.
- The System validates the input, creates a new listing in the database, and displays a success message.
- The Seller can view the newly posted listing on the marketplace page.

Exit Condition:

- The listing is successfully added and visible to all users browsing the marketplace.

Exceptions:

- E1: If the System’s database is offline, an error message is displayed.
- E2: If mandatory fields are not filled, the System prompts the Seller to complete them.

Special (Nonfunctional) Requirements:

- The listing creation must not exceed 2 seconds for confirmation (REQ-NF5).
- Data must be stored securely (REQ-NF7).

Use Case #2: Buy or Acquire a Book

Use Case Name: Buy or Acquire a Book

Participating Actors:

- Buyer (Student looking for a book)
- System

Entry Condition:

- The Buyer has accessed the website (login is optional for viewing but required for chatting).

Flow of Events:

- The Buyer visits the “Marketplace” and browses listings.
- The Buyer uses filters (course code, price range) to narrow results.
- The Buyer clicks a listing to see details.
- (Optional) If the Buyer wants more info or wants to negotiate, they must log in and use the chat function.
- The Buyer and Seller agree on a price or exchange terms, then arrange a meeting time.
- The Buyer obtains the book, and the Seller marks the listing as “sold.”

Exit Condition:

- The Buyer acquires the book.
- The listing is updated or removed from the active listings.

Exceptions:

- E1: If the item is already sold, the system notifies the Buyer and removes the listing from active search.

Special Requirements:

- Must load filtered results within 2 seconds (REQ-NF5).

Use Case #3: Create/Join Study Group

Use Case Name: Create/Join Study Group

Participating Actors:

- Student (Group creator or joiner)
- System

Entry Condition:

- The Student is logged into CCHive.

Flow of Events:

- The student navigates to “Study Groups” and clicks “Create New Group.”
- The System prompts for course/topic, description, and capacity (optional).
- The student saves the group, and the System broadcasts or lists the group for others to see.
- Other students can browse existing groups and click “Join.”
- The System records the membership and displays group info to the new member.

Exit Condition:

- The new group is visible in the system.
- Interested students have the ability to join.

Exceptions:

- E1: If the group is full, a “Group at capacity” message is shown.
- E2: If invalid data is submitted, the system shows an error.

Special Requirements:

- Real-time updates if possible (chat or push notifications).
- The system should remain stable as new groups are frequently created (REQ-NF10).

Use Case #4: Provide/Use Peer Tutoring

Use Case Name: Peer Tutoring Appointment

Participating Actors:

- Tutor (Senior student)
- Tutee (Student in need)
- System

Entry Condition:

- The Tutor is registered as a tutor on CCHive.
- The Tutee has logged in to find a tutor.

Flow of Events:

- The Tutee navigates to the tutoring section and searches by course/subject.
- The Tutee selects a Tutor profile, views availability, and requests an appointment.
- The System notifies the Tutor.
- The Tutor either accepts, declines, or proposes a new time.
- Once accepted, the System schedules the session and sends confirmations to both parties.

Exit Condition:

- The tutoring session is booked.
- Both Tutor and Tutee have a scheduled slot in their calendars.

Exceptions:

- E1: The Tutor's schedule is full; a "No available slots" message is shown.
- E2: The Tutee cancels; the session is marked canceled, and the slot reopens.

Special Requirements:

- Must handle multiple schedules without conflicts (REQ-NF9 for robust scheduling).

4.2.1 Matrix of Requirements vs. Use Cases

Use Case \ Requirement	REQ-F1 (Reg/Login)	REQ-F2 (List Book)	REQ-F3 (Filter)	REQ-F5 (Chat)	REQ-F7 (Group)	REQ-F11 (Tutor Reg.)	REQ-F12 (Book Appt.)	Others ...
UC-1 (Sell Book)	✓	✓	-	✓	-	-	-	
UC-2 (Buy Book)	Optional	-	✓	✓	-	-	-	
UC-3 (Study Group)	✓	-	-	-	✓	-	-	
UC-4 (Peer Tutoring)	✓	-	-	-	-	✓	✓	

5 Meeting Details

Date & Time: 30 January 2024, 2:00 PM

Location: College Library Study Room

Attendees:

- Harleen Kaur
- Sukhman Kaur (Leader)
- Jashanpreet Kaur
- Santa Singh
- Amritveer Singh
- Nil-Erdene Bayanburd

Discussion Points

1. **UML Modeling Tools:** The team evaluated several UML tools (Modelio, StarUML, Lucidchart) and decided to use Modelio for creating class diagrams, use case diagrams, sequence diagrams, etc.

2. **Scenarios & Use Cases:** The group created the initial set of scenarios (book selling, book buying, study group creation, peer tutoring, course rating) and drafted corresponding use cases.

3. Finalize functional and nonfunctional requirements.

4. Begin drafting UML models (use case diagrams, class diagrams, and possibly sequence diagrams).

5. Prepare for early prototype design (wireframes for key pages).

6 Team Contributions

Name	**Contribution**
Harleen Kaur (814438)	Contributed functional requirements for the marketplace; helped refine use cases for book exchange
Sukhman Kaur (Leader) (814393)	Coordinated overall report structure, wrote introduction, edited problem statement, led team meetings.
Jashanpreet Kaur (814493)	Provided UI/UX ideas; assisted in drafting the nonfunctional requirements, especially usability.
Santa Singh (814293)	Proposed login system specs, contributed to data encryption & reliability requirements.
Amritveer Singh (815257)	Researched UML diagram approaches, outlined the tutoring scenario & relevant requirements.
Nil-Erdene Bayanburd (812374)	Drafted additional scenarios, created the scenario vs. requirement matrix, and set up project timeline.